A Sharp Critical-Forcing Comparison Lemma for a Nonlinear ODE in Stochastic Convex Optimization

Solution note

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Abstract

This note proves the deterministic scalar comparison implication appearing in Conjecture 4.11 of Maulen–Soto, Fadili, and Attouch. Let

$$y'(t) \le -ay(t)^b + p(t), \qquad a > 0, \quad b > 1,$$

with $y \ge 0$ and suppose that, eventually,

$$p(t) \le P(t+1)^{-b/(b-1)}.$$

Writing

$$\alpha := \frac{1}{b-1}, \qquad \beta := \frac{b}{b-1} = \alpha + 1,$$

we prove the sharper estimate

$$\limsup_{t\to\infty} (t+1)^{\alpha} y(t) \le R_P,$$

where R_P is the largest nonnegative root of

$$aR^b - \alpha R - P = 0.$$

Consequently $y(t) = O((t+1)^{-1/(b-1)})$. Thus the note gives a solution of the scalar ODE conjecture under the standard uniform asymptotic interpretation of the hypotheses. It is only partial progress for any broader problem asking for a full stochastic localization theorem beyond this scalar comparison step, and an all-time estimate beginning at the localization time requires an additional scaled initial-data bound; without such a bound, that stronger uniform statement is false even for $p \equiv 0$.

1 Status and relation to the authors' problem

Maulen–Soto, Fadili, and Attouch reduce one part of their local stochastic convex optimization analysis to bounding nonlinear Cauchy problems of the form

$$\begin{cases} y'(t) = -ay(t)^b + p_\delta(t), & t > \widehat{t}_\delta, \\ y(\widehat{t}_\delta) = y_0(\widehat{t}_\delta, \delta) > 0, \end{cases} \tag{1}$$

where $a > 0$, $b > 1$, and p_δ is nonnegative. In Remark 4.10 and Conjecture 4.11 of their paper, the desired implication is that

$$p_\delta = O(\sigma_\infty^2), \qquad \sigma_\infty(t)^2 = O\big((t+1)^{-b/(b-1)}\big) \tag{2}$$

with constants independent of δ and $\widehat{t}_\delta$, should imply

$$y(t) = O\big((t+1)^{-1/(b-1)}\big). \tag{3}$$

The result below proves exactly this deterministic implication once the forcing estimate in (2) is granted. It does not attempt to reprove the stochastic localization estimates that produce the scalar comparison equation. Nor does it produce a closed-form formula for the solution of (1) for arbitrary p_δ; instead, it gives a sharp asymptotic comparison bound.

The conclusion is uniform in the meaningful asymptotic sense: if a common tail constant P is available in the forcing bound, then the eventual multiplicative constant in the decay estimate is independent of δ and $\widehat{t}_\delta$. A stronger estimate valid for all $t \ge \widehat{t}_\delta$ with one constant independent of the localization time is also possible, but only under a corresponding uniform scaled initial bound. Section 5 records both the positive statement and the counterexample showing that the initial bound cannot be omitted.

2 The comparison theorem

Throughout the note set

$$\alpha := \frac{1}{b-1}, \qquad \beta := \frac{b}{b-1} = \alpha + 1. \tag{4}$$

The critical exponent identity is

$$\alpha b = \alpha + 1 = \beta. \tag{5}$$

Theorem 1 (Sharp critical-forcing comparison lemma)**.** *Let $a > 0$, $b > 1$, and $t_0 \ge 0$. Let $p \in L^1_{\mathrm{loc}}([t_0,\infty))$ be nonnegative, and let $y : [t_0,\infty) \to [0,\infty)$ be absolutely continuous. Suppose that*

$$y'(t) \le -a y(t)^b + p(t) \qquad \text{for a.e. } t \ge t_0. \tag{6}$$

Assume that there are constants $P \ge 0$ and $T \ge t_0$ such that

$$p(t) \le P(t+1)^{-\beta} \qquad \text{for a.e. } t \ge T. \tag{7}$$

Let R_P be the largest nonnegative solution of

$$aR^b - \alpha R - P = 0. \tag{8}$$

Then

$$\limsup_{t\to\infty} (t+1)^\alpha y(t) \le R_P. \tag{9}$$

Consequently,

$$y(t) = O\big((t+1)^{-\alpha}\big) = O\big((t+1)^{-1/(b-1)}\big). \tag{10}$$

More explicitly, for every $\varepsilon > 0$ there is a time $T_\varepsilon \ge T$ such that

$$y(t) \le (R_P + \varepsilon)(t+1)^{-\alpha} \qquad \text{for all } t \ge T_\varepsilon. \tag{11}$$

Proof. We first record the elementary structure of the autonomous comparison field. Define

$$F(r) := \alpha r - a r^b + P, \qquad r \ge 0. \tag{12}$$

Equivalently, $-F(r) = ar^b - \alpha r - P$. Since $ar^b - \alpha r - P \to +\infty$ as $r \to \infty$ and is nonpositive at $r = 0$, the equation (8) has at least one nonnegative solution. Its largest nonnegative solution R_P satisfies

$$F(r) < 0 \quad \text{for every } r > R_P. \tag{13}$$

Indeed, if $P > 0$, then $ar^b - \alpha r - P$ has exactly one positive zero: its derivative $abr^{b-1} - \alpha$ has a single positive zero, so the polynomial first decreases and then increases, starting from the negative value $-P$. If $P = 0$, the nonnegative zeros are 0 and

$$R_0 = \left(\frac{\alpha}{a}\right)^{1/(b-1)}, \tag{14}$$

and again (13) holds for $r > R_0$.

Now make the logarithmic time change

$$s = \log(t+1), \qquad t = e^s - 1, \tag{15}$$

and introduce the scaled variable

$$u(s) := e^{\alpha s} y(e^s - 1) = (t+1)^{\alpha} y(t). \tag{16}$$

For $s \geq S := \log(T+1)$, u is absolutely continuous. By the chain rule, (6), (7), and (5), for almost every $s \geq S$,

$$\begin{aligned}
\dot{u}(s) &= \alpha e^{\alpha s} y(e^s - 1) + e^{(\alpha+1)s} y'(e^s - 1) \\
&\leq \alpha u(s) - a e^{(\alpha+1)s} y(e^s - 1)^b + e^{(\alpha+1)s} p(e^s - 1) \\
&= \alpha u(s) - a u(s)^b + e^{\beta s} p(e^s - 1) \\
&\leq \alpha u(s) - a u(s)^b + P = F(u(s)).
\end{aligned} \tag{17}$$

Thus u is a subsolution of the autonomous scalar equation

$$\dot{v} = F(v). \tag{18}$$

Let v be the solution of (18) with initial value $v(S) = u(S)$. The function F is locally Lipschitz on $[0, \infty)$, and the nonnegative half-line is forward invariant for (18), because $F(0) = P \geq 0$. Moreover, v cannot blow up in finite forward time, since $F(r) < 0$ for all sufficiently large r.

The standard one-dimensional comparison principle gives

$$u(s) \leq v(s) \qquad (s \geq S). \tag{19}$$

For completeness, here is the short proof. On any finite interval on which u and v remain bounded, choose a Lipschitz constant L for F on the relevant compact range. For $w = (u - v)_+$, the differential inequality (17) and the equation $v' = F(v)$ imply $w' \leq Lw$ for almost every time. Since $w(S) = 0$, Gronwall's inequality gives $w \equiv 0$ on that interval. Repeating on finite intervals proves (19) for all $s \geq S$.

It remains only to analyze the autonomous solution v. If $v(S) > R_P$, then $F(v) < 0$ as long as $v > R_P$, so v decreases until it approaches R_P. If $v(S) \leq R_P$, then v cannot cross above R_P, because $F(R_P) = 0$ and uniqueness prevents crossing of the equilibrium. Hence in every case

$$\limsup_{s \to \infty} v(s) \leq R_P. \tag{20}$$

Combining (16), (19), and (20) gives (9). The big-O estimate (10) follows immediately. Finally, (11) is just the definition of the limsup bound applied to $u(s) = (t+1)^{\alpha} y(t)$. $\square$

Remark 1 (No equality is needed)**.** *The authors' scalar Cauchy problem is an equality. The theorem is stated for a differential inequality because this is the natural form in comparison arguments. Any solution of*

$$y' = -ay^b + p(t)$$

automatically satisfies (6), so the theorem applies directly to the equality case.

3 Application to Conjecture 4.11

Corollary 1 (Uniform family version)**.** *Let D be an index set. For every $\delta \in D$, let $\widehat{t}_\delta \geq 0$, let $p_\delta \geq 0$ be locally integrable on $[\widehat{t}_\delta, \infty)$, and let $y_\delta : [\widehat{t}_\delta, \infty) \to [0, \infty)$ be absolutely continuous with*

$$y_\delta'(t) \leq -ay_\delta(t)^b + p_\delta(t) \qquad \text{for a.e. } t \geq \widehat{t}_\delta. \tag{21}$$

Assume that there is a constant $P \geq 0$, independent of δ and $\widehat{t}_\delta$, such that for each δ there exists a tail time $T_\delta \geq \widehat{t}_\delta$ with

$$p_\delta(t) \leq P(t+1)^{-\beta} \qquad \text{for a.e. } t \geq T_\delta. \tag{22}$$

Then

$$\limsup_{t\to\infty} (t+1)^\alpha y_\delta(t) \leq R_P \qquad (\delta \in D), \tag{23}$$

where R_P is the largest nonnegative root of (8). Thus, for every fixed $\varepsilon > 0$ and every δ,

$$y_\delta(t) \leq (R_P + \varepsilon)(t+1)^{-\alpha} \tag{24}$$

eventually in t, with multiplicative constant independent of δ and $\widehat{t}_\delta$.

Proof. Apply Theorem 1 to each member of the family, starting at its own tail time T_δ. The same value of P gives the same root R_P for every δ. □

Corollary 2 (The scalar conjecture)**.** *Suppose that the family of Cauchy problems (1) satisfies*

$$p_\delta(t) = O(\sigma_\infty(t)^2), \qquad \sigma_\infty(t)^2 = O\big((t+1)^{-\beta}\big), \tag{25}$$

under the standard uniform interpretation that the multiplicative constants in the tail estimates can be chosen independently of δ and $\widehat{t}_\delta$. Then

$$y_\delta(t) = O\big((t+1)^{-\alpha}\big) = O\big((t+1)^{-1/(b-1)}\big) \tag{26}$$

for each δ, with the eventual multiplicative constant independent of δ and $\widehat{t}_\delta$.

Proof. The hypotheses (25) imply that, after passing to a sufficiently late tail time if necessary, there is a constant $P \geq 0$ independent of δ and $\widehat{t}_\delta$ such that

$$p_\delta(t) \leq P(t+1)^{-\beta}.$$

Corollary 1 then gives (26). □

4 Sharpness

The exponent and the root constant in Theorem 1 are sharp for the critical power forcing. Fix $P \geq 0$ and set

$$p(t) = P(t+1)^{-\beta}. \tag{27}$$

For a constant $C \geq 0$, the power profile

$$y(t) = C(t+1)^{-\alpha} \tag{28}$$

is an exact solution of

$$y' = -ay^b + P(t+1)^{-\beta} \tag{29}$$

if and only if

$$aC^b - \alpha C - P = 0. \tag{30}$$

Indeed,

$$y'(t) = -\alpha C(t+1)^{-\beta},$$

while

$$-ay(t)^b + P(t+1)^{-\beta} = (-aC^b + P)(t+1)^{-\beta},$$

and these expressions agree exactly when (30) holds. Taking $C = R_P$ gives an exact solution with

$$\lim_{t\to\infty} (t+1)^\alpha y(t) = R_P.$$

Therefore no theorem using only the tail coefficient P can replace R_P by a smaller universal limsup constant.

5 Uniformity at the localization time

The preceding results give a uniform eventual multiplicative constant, but not a uniform estimate from the localization time onward. The distinction is essential.

Proposition 1 (All-time bound under a scaled initial envelope)**.** *In the setting of Corollary 1, assume in addition that the forcing bound holds from the localization time itself,*

$$p_\delta(t) \leq P(t+1)^{-\beta} \qquad \textit{for a.e. } t \geq \widehat{t}_\delta, \tag{31}$$

and that there is a constant $Y \geq 0$, independent of δ, such that

$$(\widehat{t}_\delta + 1)^\alpha y_\delta(\widehat{t}_\delta) \leq Y. \tag{32}$$

Then

$$y_\delta(t) \leq \max\{Y, R_P\}(t+1)^{-\alpha} \qquad \textit{for every } t \geq \widehat{t}_\delta, \tag{33}$$

with a constant independent of δ and $\widehat{t}_\delta$.

Proof. Repeat the logarithmic scaling proof from Theorem 1, now starting at

$$S_\delta = \log(\widehat{t}_\delta + 1).$$

The scaled variable

$$u_\delta(s) = e^{\alpha s} y_\delta(e^s - 1)$$

satisfies $\dot{u}_\delta \leq F(u_\delta)$ for $s \geq S_\delta$, and (32) gives $u_\delta(S_\delta) \leq Y$. Since $F(r) < 0$ for $r > R_P$, the interval $[0, \max\{Y, R_P\}]$ is forward invariant for the comparison equation $v' = F(v)$. Comparison therefore yields

$$u_\delta(s) \leq \max\{Y, R_P\} \qquad (s \geq S_\delta),$$

which is equivalent to (33). □

Proposition 2 (The scaled initial envelope is necessary)**.** *Without a uniform bound of the form* (32)*, one cannot in general have a constant C such that*

$$y_\delta(t) \leq C(t+1)^{-\alpha} \qquad \textit{for all } t \geq \widehat{t}_\delta \tag{34}$$

for every member of the family, even when $p_\delta \equiv 0$.

Proof. For each integer $N \geq 1$, set $\widehat{t}_N = N$, $p_N \equiv 0$, and choose the initial value

$$y_N(N) = 1.$$

This gives a legitimate solution of $y' = -ay^b$ on $[N, \infty)$. If (34) held with a single constant C independent of N, then evaluating at $t = N$ would give

$$1 \leq C(N+1)^{-\alpha}.$$

This is impossible for all sufficiently large N, since $\alpha > 0$. Thus some uniform scaled control of the initial data is necessary for an all-time estimate beginning at the localization time. □

6 Conclusion

The proof above gives a solution of the scalar deterministic comparison problem in Conjecture 4.11: critical forcing of order $(t+1)^{-b/(b-1)}$ implies solution decay of order $(t+1)^{-1/(b-1)}$, with a sharp limsup constant determined by the largest root of $aR^b - \alpha R - P = 0$. The result should be viewed as partial progress only if the intended target is broader than this scalar comparison implication, for example a full stochastic theorem establishing the forcing estimate itself, or a uniform all-time estimate from $\widehat{t}_\delta$ without any scaled initial-data hypothesis. The latter stronger statement is false in general by Proposition 2.

References

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